NATIONAL GDP GROWTH PREDICTOR

INTRODUCTION

Economic growth is one of the most important indicators of a country's development. A strong economy leads to better employment opportunities, improved living standards, higher income levels, and overall social progress. One of the key measures of economic performance is **GDP per capita**, which represents the average income earned per person in a country.

In the modern digital era, internet access has become a critical factor influencing economic activities. Increased internet penetration enhances communication, enables digital businesses, improves access to education and financial services, and supports innovation. Countries investing in digital infrastructure often experience improvements in productivity and economic performance.

Under **Sustainable Development Goal (SDG) 8 – Decent Work and Economic Growth**, promoting inclusive and sustainable economic growth is a global priority. Digital connectivity plays a significant role in achieving this goal by enabling entrepreneurship, employment generation, and global trade participation.

This project aims to analyze and model the relationship between a country's internet penetration rate and its GDP per capita using time-series analysis and machine learning techniques. By studying historical economic data, the project attempts to understand whether increasing digital connectivity contributes to national economic growth.

PROBLEM STATEMENT

Despite rapid global digital transformation, the quantitative impact of internet penetration on a country's economic growth remains an important area of analysis. Economic data is often volatile and influenced by multiple external factors such as inflation, global crises, and policy changes. Therefore, understanding the true relationship between internet usage and GDP growth requires proper time-series alignment, noise reduction, and predictive modeling.

The problem addressed in this project is:

To model and predict GDP per capita using internet penetration rate as a key explanatory variable by applying time-series preprocessing techniques and machine learning models.

The project focuses on:

- Aligning multi-year economic datasets chronologically
- Removing noise from volatile economic indicators
- Building predictive models to estimate GDP growth
- Evaluating whether internet penetration significantly influences economic performance

The ultimate objective is to provide data-driven insights into how digital infrastructure development can support sustainable economic growth under SDG 8.

METHODOLOGY

The methodology adopted for the project *"National GDP Growth Predictor"* follows a systematic approach to analyze the relationship between internet penetration and GDP per capita using time-series data and machine learning techniques.

Data Collection

Economic data was collected from reliable international databases such as:

- World Bank
- International Monetary Fund

The following indicators were considered:

- GDP per capita (current US$)
- GDP growth rate (annual %)
- Internet penetration rate (% of population)
- Year and Country

The dataset includes multi-year time-series data to observe long-term trends.

Data Preprocessing

Since economic data is often inconsistent and incomplete, preprocessing was performed to ensure data quality.

✓ Handling Missing Values

- Removed records with excessive missing data
- Applied interpolation where appropriate

✓ Data Type Conversion

- Converted numerical columns to appropriate formats
- Ensured year column was properly ordered

✓ Time-Series Alignment

- Aligned GDP and internet penetration data chronologically
- Ensured matching years across variables
- Sorted dataset in ascending order by year

This step prevents incorrect correlation caused by misaligned data.

Noise Removal and Smoothing

Economic indicators are volatile due to global events (pandemics, financial crises, policy changes).

To reduce short-term fluctuations:

- Applied Rolling Mean (3-year moving average)
- Used smoothing techniques to identify long-term trends

This improves model stability and prediction accuracy.

Exploratory Data Analysis (EDA)

To understand the data patterns, the following analyses were performed:

- Line plots to observe GDP and internet growth trends
- Scatter plots to visualize relationship between variables
- Correlation coefficient calculation
- Trend analysis over years

EDA helps determine whether the relationship is linear or non-linear.

Feature Engineering

To improve model performance, additional features were created:

- Lag features (Internet penetration of previous year)
- Growth rate percentage
- Smoothed internet penetration values

Lag features are important because economic impact is not immediate and may appear after a certain time period.

Model Development

The following machine learning models were implemented:

◆ **Linear Regression**

Used as a baseline model to understand linear relationship.

◆ **Random Forest Regression**

Used to capture non-linear relationships and improve prediction accuracy.

◆ **LSTM (Advanced Model)**

Long Short-Term Memory networks were explored to handle sequential time-series data effectively.

Model Evaluation

The performance of the models was evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- R^2 Score

These metrics help measure prediction accuracy and model reliability.

Interpretation and Conclusion

The final step involved analyzing model outputs and interpreting results in economic terms. The study examined whether higher internet penetration significantly contributes to GDP growth and how digital infrastructure investment can support sustainable development under SDG 8.

Summary of Methodology Flow

Data Collection → Data Cleaning → Time-Series Alignment → Noise Removal → EDA → Feature Engineering → Model Building → Evaluation → Interpretation

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